

HANKO RUSSARO, FinlandWMO: **029820**Lat: **59.774N**Lon: **22.952E**Elev: **7**StdP: **101.25**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.4	-12.5	-17.7	0.8	-15.0	-14.9	1.0	-12.1	19.9	-2.3	18.2	-0.6	7.0	50	0.803

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	23.6	19.8	22.2	18.8	21.0	18.0	20.6	22.8	19.5	21.6	18.5	20.5	6.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	14.4	22.1	18.6	13.5	21.1	17.5	12.6	19.9	59.2	22.9	55.7	21.8	52.2	20.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	16.3	14.7	-16.5	25.2	4.0	2.3	-19.3	26.9	-21.7	28.2	-23.9	29.5	-26.9	31.2		
(5)			-16.8	20.9	3.9	1.6	-19.6	22.0	-21.9	22.9	-24.1	23.8	-26.9	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-1.9	-2.8	-0.8	3.3	8.4	13.4	17.5	17.3	13.4	8.0	3.8	0.8
	DBStd	7.79	4.49	4.92	3.32	2.25	2.84	2.43	2.54	2.23	2.47	3.16	3.47	4.04
	HDD10.0	1879	368	359	334	201	66	2	0	0	4	75	186	286
	HDD18.3	4258	626	593	592	451	307	148	47	46	149	319	436	544
	CDD10.0	702	0	0	0	0	17	105	233	228	104	14	0	0
	CDD18.3	38	0	0	0	0	0	1	22	15	0	0	0	0
	CDH23.3	49	0	0	0	0	0	1	39	9	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	7.4	8.3	7.9	7.5	6.6	6.3	6.4	6.0	6.3	7.3	8.2	8.8	9.2
Precipitation	PrecAvg	657	61	43	40	35	29	41	57	75	55	76	80	66
	PrecMax	762	108	98	93	63	62	77	166	176	145	177	163	120
	PrecMin	478	8	3	5	9	5	14	1	7	15	30	7	19
	PrecStd	81	27	26	19	15	16	18	43	41	33	35	39	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	4.1	6.1	11.6	19.1	21.9	26.1	24.5	19.9	14.7	10.3	8.1
		MCWB	4.9	3.1	3.7	7.6	13.7	16.7	20.7	20.1	16.6	13.0	9.4	7.2
	2%	DB	4.7	3.2	4.5	9.2	16.2	19.7	24.4	23.0	18.4	13.7	9.7	7.2
		MCWB	4.1	2.5	2.9	6.3	12.3	15.6	20.4	19.3	15.9	12.4	8.9	6.4
	5%	DB	3.9	2.7	3.5	7.7	14.2	18.4	23.0	21.8	17.5	12.9	8.9	6.2
		MCWB	3.2	2.1	2.3	5.3	10.8	15.0	19.4	18.6	15.3	11.9	8.1	5.4
	10%	DB	3.1	2.2	2.7	6.4	12.7	17.2	21.7	20.7	16.6	12.1	8.1	5.4
		MCWB	2.4	1.7	1.7	4.5	9.8	14.4	18.7	17.9	14.7	11.1	7.2	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	3.5	4.1	8.1	14.5	17.7	22.3	21.4	17.6	13.5	9.9	7.5
		MCDB	5.5	3.8	5.5	10.6	18.0	20.2	24.4	23.6	19.1	14.1	10.1	7.8
	2%	WB	4.2	2.7	3.2	6.7	12.9	16.5	21.1	20.2	16.5	12.7	9.1	6.6
		MCDB	4.6	3.0	4.1	8.8	15.5	18.8	23.6	22.2	17.8	13.4	9.5	7.1
	5%	WB	3.4	2.2	2.5	5.7	11.4	15.5	20.1	19.2	15.7	12.0	8.3	5.6
		MCDB	3.8	2.6	3.3	7.3	13.6	17.7	22.4	21.1	16.9	12.8	8.9	6.2
	10%	WB	2.6	1.8	2.0	4.8	10.2	14.7	19.1	18.4	15.1	11.3	7.3	4.7
		MCDB	3.0	2.2	2.6	6.2	12.2	16.7	21.3	20.1	16.4	12.0	8.0	5.3
Mean Daily Temperature Range	5% DB	MDBR	3.4	3.5	3.9	4.0	4.6	4.4	4.5	4.2	3.7	3.0	2.8	3.0
		MCDBR	2.6	2.7	3.7	5.8	7.0	6.1	5.9	5.0	4.3	3.0	2.6	2.5
		MCWBR	2.7	2.5	2.8	3.7	3.9	3.3	3.1	2.9	3.0	2.7	2.9	2.6
	5% WB	MCDBR	2.7	2.5	3.3	5.3	6.0	5.2	5.1	4.2	3.8	2.7	2.6	2.5
		MCWBR	2.9	2.5	2.7	3.6	3.8	3.3	3.0	3.0	2.9	2.8	3.0	2.7
Clear-Sky Solar Irradiance	taub	0.283	0.315	0.335	0.372	0.367	0.363	0.391	0.383	0.353	0.336	0.313	0.269	
	taud	2.167	2.232	2.296	2.321	2.419	2.473	2.419	2.428	2.496	2.404	2.218	2.101	
	Ebn at Noon	489	680	794	826	861	870	833	810	778	660	445	350	
	Edh at Noon	44	71	90	106	103	99	103	95	75	61	42	31	
All-Sky Solar Radiation	RadAvg	0.24	0.76	2.29	3.96	5.56	5.88	5.55	4.37	2.66	1.15	0.31	0.15	
	RadStd	0.04	0.14	0.27	0.41	0.54	0.37	0.47	0.43	0.33	0.15	0.06	0.02	